INNOVATION CASE STUDY · MARCH 2026

AI-Powered Procurement Document Automation

How a Custom AI Agent Transforms Philippine Government Procurement Document Preparation from Days to Minutes

Multi-day manual document prep → sub-10-minute AI-assisted conversation

n8n · GPT-5.2 · Gemini Flash · Flask · Google Drive

Executive Summary

THE PROBLEM

Procurement officers spend hours or days manually entering the same project data into 8 required government templates — with arithmetic errors that risk voiding the entire procurement.

THE SOLUTION

A 3-module AI pipeline: a conversational GPT-5.2 agent, Gemini Flash vision extraction, and a Flask document renderer — delivering a ready-to-use Google Drive folder link.

THE RESULT

All 8 required procurement documents generated in under 10 minutes. Zero manual data entry. Zero arithmetic errors. One shareable Drive link.

THE IMPACT

Procurement officers freed from mechanical document work — enabling focus on project oversight, stakeholder coordination, and public accountability.

The Problem

RA 9184 mandates a specific set of documents for every infrastructure project — DepEd, DPWH, and LGUs face this burden on every procurement cycle.

01

Manual Document Duplication

The same project data entered by hand into 8+ templates — multiplying inconsistency risk across every document.

02

Arithmetic Errors

Unit costs, labor additions, and totals computed manually with no programmatic validation — errors invite audit findings or void procurements entirely.

03

Slow Turnaround

Officers wait days for drafted documents from consultants, stalling entire project timelines and delaying disbursements.

System Architecture

Three-module AI pipeline: conversation · calculation · generation

MODULE 1

Chat Pipeline

/procurement-chat

Conversational AI entry point

Accepts text, images, or PDFs. Routes input, runs Gemini vision extraction if needed, and drives GPT-5.2 agent conversation with full session memory.

MODULE 2

Calculator Service

/procurement-calculator

Deterministic math engine

Computes material totals, labor costs, and grand totals. Scales all quantities proportionally when a target budget differs from the computed total — no AI arithmetic.

MODULE 3

Document Generator

/procurement-generate

File production & cloud delivery

Flask API renders all 8 procurement documents. n8n creates a named Google Drive folder, uploads each file, and returns a direct shareable folder URL.

Six Innovation Dimensions

01 Conversational AI Interface

GPT-5.2 acts as a Philippine procurement expert — not a generic chatbot. Derives 'DepEd' from a school name; never re-asks what it already knows.

02 Multimodal Vision Extraction

Accepts photo or PDF scan. Gemini Flash extracts all structured data — bidder details, line items, quantities, unit costs — returning JSON within seconds.

03 Dedicated Calculation Engine

AI is explicitly prohibited from computing math. A deterministic JavaScript engine handles all arithmetic — eliminating LLM hallucination from the critical path.

04 Stateful Session Memory

20-message context buffer per session ID. Users can correct earlier data, build info incrementally, and reference prior statements — no restarting.

05 Intelligent Payload Normalization

Four-pass JSON parsing handles inch marks, curly quotes, non-ASCII chars, and malformed structures. Field name variants all resolve to the same canonical field.

06 Automated Cloud Delivery

A single tool call triggers Flask to render all 8 documents. n8n creates a named Drive folder, uploads each file, and returns a shareable URL with file count.

Results at a Glance

8	<10m	3	0
Documents Generated	End-to-End Turnaround	AI Models Integrated	Arithmetic Errors

TIME COMPARISON

AI: ~9 min

Manual: ~480 min (8 hrs)

Lessons Learned & Version 2.0

WHAT WE LEARNED

AI as domain expert — deep procurement prompts are the key differentiator in GovTech

Separate AI judgment from arithmetic — routing math to deterministic code eliminates hallucination risk

Multimodal input is a force multiplier — photo ingestion removes the data entry barrier for legacy workflows

Resilience engineering compounds value — multi-pass parsing separates a prototype from production

VERSION 2.0 ROADMAP

- Web-based front-end portal — eliminate webhook complexity for non-technical officers
- Expand document set to full RA 9184 package including BAC resolutions and Notice of Award
- Audit trail logging — timestamped record of every field's source and generating session
- Multi-project batch mode — process multiple datasets in a single session
- Localize for ARMM, Visayas, and Mindanao regulatory variants

Key Insights

AI as Domain Expert

Configure AI with deep domain knowledge — not generic prompts. The procurement expertise embedded in the system prompt is what makes the interface practical.

Hybrid Intelligence Model

AI for judgment and language understanding; deterministic code for precision calculations. This hybrid is the correct model for any accuracy-critical workflow.

Multimodal = Backward Compatible

Vision-based ingestion means the system works with legacy paper processes — dramatically expanding the addressable user base beyond digital-native offices.

GovTech Blueprint

The pattern — conversational intake → vision ingestion → deterministic calculation → automated generation → cloud delivery — is replicable across Philippine regulatory workflows.

The most valuable AI applications in government are not those that replace human judgment — but those that eliminate mechanical, repetitive work.

By automating document generation without sacrificing accuracy or compliance, the system frees procurement officers to focus on what matters: project oversight, stakeholder coordination, and public accountability.

n8n · OpenAI GPT-5.2 · Google Gemini Flash · Flask · Google Drive · March 2026